

Critical error rate in practice: context-aware variant significance for nanopore amplicon pipelines

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Abstract

The critical error rate (CER) framework [1] provides a per-variant significance statistic p^* that is interpretable against known platform error rates. The framework as originally posed assumes a single context-free error rate, which is appropriate for non-homopolymer substitutions in nanopore data but breaks down for homopolymer (HP) length variants, where empirical error rates vary by a factor of two or more across HP length and base context [2]. We present a context-aware extension of the framework, retaining the single binomial equation but replacing the uniform error rate p with a context-specific q_{ctx} derived from empirical measurement. To preserve cross-variant comparability under context-aware q_{ctx} , we introduce the *CER factor*, p^*/q_{ctx} , as the primary derived metric. The factor is dimensionless, comparable across variants and specimens, degrades linearly under q_{ctx} misspecification, and re-scales cleanly when the empirical error table is updated. We describe the implementation in the speconsense pipeline, including N-denominator choice for post-phasing validation, identity-group construction, pairwise CER comparison, and a metadata reproducibility design that allows stored output to be re-evaluated against revised assumptions without re-running the pipeline.

1 Introduction

The critical error rate (CER) framework, introduced in [1], asks the inverse of the conventional p-value question: rather than testing whether an observed variant is significant under an assumed error rate, it derives the per-position error rate p^* at which the observed pattern would become plausible as artifact. A practitioner then compares p^* against their platform’s known error rate; well-supported variants produce p^* values several times the platform rate, providing a transparent margin for judgment.

The framework’s strength is that p^* has the same units as the platform error rate the reader already knows, decoupling the analysis from the moving target of basecaller and chemistry improvements. A p^* of 6.8% is robust against any reasonable nanopore configuration without committing to a specific assumed rate at output time.

1.1 The context-dependence problem

The framework as originally formulated assumes a single per-position error rate p characterizes the platform. For non-homopolymer substitutions in current nanopore chemistry, this is approximately true: empirical post-clustering rates are tightly clustered around 0.6%–1.0% across position contexts [2]. The framework’s $p^* = 6.8\%$ headline against an assumed rate of 1.5% is robust to this spread.

For homopolymer-length variants, the assumption breaks down. Homopolymer (HP) length errors are the dominant systematic error mode in nanopore sequencing [4], and their empirical rates depend strongly on context:

- HP length error rate increases monotonically from $\sim 0.7\%$ per position at $L = 1$ to $\sim 1.5\%$ at $L = 5$, and steeply thereafter [2].
- G/C homopolymers consistently produce higher error rates than A/T at matched length, with the gap widening at longer lengths.
- Deletion-biased errors dominate insertion-biased errors at a 3–5:1 ratio across moderate HP lengths.
- Per-run and per-basecaller variation contribute additional spread of $\sim 0.6\times$ to $2.2\times$ on top of the context dependence.

A single assumed error rate cannot serve all of these contexts simultaneously without either over-suppressing variants in low-error contexts (e.g., $L = 1$ HP positions) or under-suppressing in high-error contexts (e.g., $L = 5$ G runs). The current speconsense default of 1.5% assumed error rate is calibrated to cover the non-HP substitution regime; applied uniformly to HP variants, it loses 30–50% of the available signal.

1.2 Contributions

This paper extends the CER framework to context-dependent error rates without abandoning its core appeal:

1. A **unified equation** that retains the single binomial structure of [1] but replaces L with a count of independent test sites n_{sites} and replaces q with a context-specific q_{ctx} . For multi-position variants ($K > 1$), the joint per-read error probability becomes $\prod_i q_{\text{ctx},i}$ instead of q^K , accommodating mixed contexts naturally.
2. A **practical decision metric**, the *CER factor*, defined as p^*/q_{ctx} . The factor is dimensionless and context-comparable by construction, recovering the cross-variant comparability that raw p^* loses under context-aware q_{ctx} . We compare it against alternative metrics (raw p^* , p-value) and recommend it as the primary annotation.
3. Implementation choices in the **speconsense pipeline** that follow from the framework: how N is chosen for post-phasing validation, how identity groups define the comparison set for pairwise CER, why phasing deliberately operates without statistical gating, how q_{ctx} is parameterized, and how HP detection is reordered relative to homopolymer normalization.
4. A **metadata reproducibility design** that stores enough information in each specimen’s output to recompute p^* , p-value, and CER factor under revised assumptions, without re-running the upstream pipeline.

This paper assumes the reader is familiar with the theoretical CER framework of [1] and the empirical HP error characterization of [2]. Section references to these companion documents are made where useful but the core results are not re-derived.

2 Unified equation

We retain the binomial structure of the CER paper [1]. The generalization replaces the amplicon length L with a count of independent test sites n_{sites} and replaces the single per-position error probability q with a context-specific q_{ctx} .

2.1 Single-position variants ($K = 1$)

For a candidate variant supported by M reads at a single position with context c in a specimen of N total reads,

$$n_{\text{sites}} \cdot P(\text{Binom}(N, q_{\text{ctx},c}) \geq M) = T, \quad (1)$$

where T is either an input significance level α (when solving for p^*) or an output p-value (when $q_{\text{ctx},c}$ is fixed and the equation is evaluated). This reduces to the formulation in [1] when $q_{\text{ctx},c} = q$ is uniform across positions and $n_{\text{sites}} = L$.

2.2 Multi-position variants ($K > 1$)

When a variant is distinguished from its comparison cluster at $K > 1$ positions, the joint per-read probability of erroring at all K positions under the null is the product of per-position rates:

$$\binom{n_{\text{sites}}}{K} \cdot P(\text{Binom}(N, \prod_{i=1}^K q_{\text{ctx},i}) \geq M) = T. \quad (2)$$

This generalization is essential for HP applications because mixed-context $K > 1$ variants are common: a multi-position variant may be distinguished at one HP position and one non-HP position simultaneously. Equation 2 reduces to $\binom{L}{K} \cdot P(\text{Binom}(N, q^K) \geq M) = T$ when all $q_{\text{ctx},i}$ are equal.

2.3 Bonferroni site counting

The site count n_{sites} is the number of independent positions tested. Three definitions are operationally distinct:

Per-base count. $n_{\text{sites}} = L$, the amplicon length in bases. This is the formulation in [1]; appropriate when every position is tested independently.

Per-site count with HP aggregation. $n_{\text{sites}} = (\text{non-HP positions}) + (\text{HP runs of } L \geq 2)$. A length-5 HP run is one site for length-variant testing, not five. For ITS amplicons of ~ 700 bp, this aggregation reduces n_{sites} modestly (typically by ~ 15 – 25%) because most HP runs are short.

Per-site count with alternative-base correction. Each non-HP position admits 3 alternative bases; each $L \geq 2$ HP run admits multiple alternative lengths. A strict Bonferroni would multiply n_{sites} by the number of plausible alternatives per site. Following [1], we treat the $q \rightarrow q/3$ uniform error split as implicit per-position correction over alternative bases for non-HP positions, and use empirical per-transition rates for HP positions (which are already specific to one alternative).

We adopt the per-site count with HP aggregation as the default. The choice affects p^* only logarithmically (a 25% reduction in n_{sites} shifts p^* by less than 5%).

3 The CER factor

3.1 Definition

The *CER factor* is the dimensionless ratio

$$\text{CER factor} = \frac{p^*}{q_{\text{ctx}}}. \quad (3)$$

It answers the question: *by what factor would the platform error rate at this variant’s context need to be inflated to make the observation plausible as artifact?* A factor of 4 means quadruple the empirical error rate would be needed; a factor of 1 corresponds exactly to the platform rate.

3.2 Comparison with alternative metrics

Three derived metrics summarize the same underlying CER surface:

Raw p^* . The original CER paper’s headline metric. Interpretable as “the per-position error rate that would explain this variant.” Comparable across variants only when q_{ctx} is approximately uniform.

p-value. The conventional statistic, computed by fixing q_{ctx} and evaluating Equation 1 for T . Universal 0–1 scale, composable for FDR and meta-analysis, exponentially sensitive to q_{ctx} misspecification.

CER factor. Defined by Equation 3. Dimensionless, context-independent on the output side, linearly sensitive to q_{ctx} misspecification, re-scalable when q_{ctx} updates.

Table 1 summarizes the trade-offs.

Table 1: Trade-off summary for derived CER metrics. “✓” indicates the metric is preferred on that dimension.

Property	Raw p^*	p-value	CER factor
Context-comparable across variants		✓	✓
Linear sensitivity to q_{ctx} error			✓
Composable for FDR / meta-analysis		✓	
Principled default decision threshold		✓	
Reinterpretable against future chemistry	✓		✓
Familiar to general statistical audience		✓	
Bounded dynamic range at strong end			✓

3.3 Why the factor is the right primary metric for amplicon pipelines

For speconsense’s primary use case — amplicon barcoding where consensus sequences are deposited into taxonomic reference databases — the CER factor’s properties align with what practitioners need:

1. **Cross-variant comparability is the dominant requirement.** Validators rank variants by confidence and apply a threshold. The factor provides a single comparable scale across HP and non-HP contexts.
2. **Linear sensitivity to q_{ctx} is honest.** Empirical error rates vary $\sim 0.6\times$ between sequencing runs on the same basecaller [2]. P-values would swing by orders of magnitude under this drift; the factor moves linearly, which more accurately reflects our uncertainty about the true error rate at any given site.
3. **Re-scaling under updated chemistry is direct.** If a future q_{ctx} table revises the rate at a context by a factor r , stored factors at that context can be rescaled by $1/r$ without recomputing from raw N and M . P-values must be recomputed from raw inputs; raw p^* does not require rescaling but loses cross-context comparability.

4. **The robustness interpretation survives.** A factor of 4 is “quadruple the empirical floor would be needed to explain this as artifact” — a direct generalization of the raw p^* interpretation, retaining the human-interpretable units that motivated CER in the first place.

3.4 Default decision threshold

We propose a default factor threshold of ≥ 2 for accepting a variant: the observed pattern requires at least double the empirical context-specific error rate to be explained as artifact. The threshold is conservative under typical q_{ctx} uncertainty (run-to-run variation, basecaller drift) and recovers the boundary behavior of the existing default $p^* \geq 1.5\%$ against a non-HP $q_{\text{ctx}} \approx 0.7\%$ (factor ≈ 2.1).

The threshold is a tunable: stricter calls (factor ≥ 3) are appropriate for novel reference deposits; permissive calls (factor ≥ 1) are appropriate for diversity surveys where sensitivity matters more than specificity. Section 6 describes how the factor can be recomputed at any threshold from stored metadata without re-running the pipeline.

3.5 FASTA annotation scheme

Two CER fields are emitted per consensus header. The factor is the primary decision metric; the details field is a structured string that gives the reader enough context to interpret the factor without reproducing the full metadata (which lives in the per-specimen JSON, Section 6).

```
>sample-c1 size=200 ric=100 rid=95.2 cer_factor=4.2 cer_details=p*=0.031;
  ctx=non-hp-sub;q=0.0059
```

For mixed-context $K > 1$ variants, the details field encodes per-position context tags and rates joined by +, with the explicit K value:

```
>sample-c2 size=42 ric=42 rid=98.7 cer_factor=246.6 cer_details=p*=0.031;K
  =2;ctx=non-hp-sub+hp-13-A-del1;q=0.0059+0.0213
```

The two fields:

- **cer_factor**: the headline decision metric, p^*/q_{ctx} . A single dimensionless number; higher means more robust.
- **cer_details**: a semicolon-separated structured string encoding p^* , the K value (omitted when $K = 1$), the per-position context tag(s), and the per-position empirical error rate(s). Per-position values for $K > 1$ are joined by + in the order the positions were evaluated. Provides enough information to interpret **cer_factor** (e.g., distinguish a single-position non-HP variant from a multi-position mixed-context variant) and to sanity-check the factor’s derivation against a remembered platform rate. The full per-position breakdown, α , N , M , n_{sites} , identity-group membership, and recomputation inputs live in the per-specimen metadata JSON (Section 6); the header carries only what a reader needs to triage a variant at a glance.

4 Parameterizing q_{ctx}

The unified framework demands a context classifier and a q_{ctx} table. This section describes the context taxonomy, the source of empirical values, and the operational choices for keeping q_{ctx} current.

4.1 Context classes

Each MSA position falls into one of a small set of context classes. The classification is deterministic from the consensus sequence at that position and depends on no upstream call:

Non-HP substitution. The consensus base is not part of an HP run of length ≥ 2 , and the variant differs by a substitution. Empirical q_{ctx} corresponds to the per-position substitution rate; for Dorado v5.0 on R10.4.1, $q_{\text{ctx}} \approx 0.0059$ [2].

HP-interface substitution. The consensus base is the first or last base of an HP run of length ≥ 2 , and the variant is a substitution. Empirical analysis [2] §4 demonstrates that interface substitutions occur at rates indistinguishable from non-HP substitutions; they are assigned the non-HP substitution rate.

HP length change at (b, L, δ) . The variant differs from the consensus by a length change in an HP run of base b at length L by amount δ (e.g., $\delta = -1$ for a one-base deletion). Empirical q_{ctx} is read from the per-(base, length, direction) table in [2] §3.3.

HP length change at $L \geq 6$. Suppressed by blanket normalization; no q_{ctx} assigned and no CER evaluation. This follows the recommendation of [2] §8.1: at $L \geq 6$ accuracy degrades sharply and variant calling becomes unreliable.

Indel at non-HP position. Treated as a substitution-class event with the non-HP rate as a placeholder. Indels at non-HP positions are rare in amplicon data and do not warrant a separate empirical column at this stage.

The classifier returns a context tag and a q_{ctx} value. For mixed- context $K > 1$ variants, the per-position tags are recorded as a list and the joint q_{ctx} is the product (Equation 2).

4.2 Sources of q_{ctx} values

Three operational regimes for q_{ctx} are appropriate at different levels of pipeline maturity:

Phase 1 (initial deployment): shipped per-basecaller table. A static q_{ctx} table is bundled with speconsense, parameterized by basecaller identifier (e.g., `dorado-v5.0.0`). Values are drawn from the empirical analysis in [2] for each (base, length, direction) combination. The table includes a version stamp recorded in metadata (Section 6). Users select their basecaller via a CLI flag (`--qctx-table dorado-v5.0.0`) or accept the default.

Phase 2 (mature deployment): per-run estimation. q_{ctx} is estimated from the current run’s MSAs at pipeline startup. The estimator pools positions across all specimens after applying the outlier-and-bimodal-detection biological-variant filter of [2] §5.3. The estimated q_{ctx} is recorded in metadata so that downstream interpretation can use the run-specific values or fall back to the shipped defaults.

Override: user-supplied table. A flag (`--qctx-table-file PATH`) accepts a TSV with columns (`context_tag`, `q_ctx`). Useful for research applications that need to characterize variant calls under synthetic or extreme error assumptions.

The first deployment supports Phase 1 only. Phase 2 is a substantial implementation effort and is left for future work; it is mentioned here so that the metadata schema (Section 6) is forward-compatible.

4.3 Stability across runs

The cross-run scaling factor between ont98 and run116 (same basecaller, three months apart) is $\sim 0.6\times$ at $L = 1-5$ [2] §5.4. This means a shipped Phase 1 table may differ from a specific run’s actual rates by up to a factor of 2. The CER factor’s linear sensitivity makes this a roughly $2\times$ shift in factor values, which is large but interpretable; p-values under the same misspecification would shift by orders of magnitude. Phase 2 per-run estimation eliminates this source of uncertainty.

5 Implementation in speconsense

This section describes how the unified framework integrates with the existing speconsense pipeline. The current implementation [3] performs recursive MSA-based variant phasing, post-phasing identity grouping, and pairwise CER validation. The context-aware framework requires changes at each of these stages.

5.1 N denominator choice

CER is evaluated only at post-phasing pairwise validation (Section 5.4); phasing itself uses local frequency and count thresholds without statistical gating. The denominator N in the binomial therefore has a single definition at the only point where it enters the computation.

Identity-group sum. When validating one cluster against another within an identity group, N is the sum of read counts within the group, not the total specimen count. The pairwise comparison’s null hypothesis is “these two clusters represent the same underlying sequence, with the smaller cluster arising as artifact of the larger”—a question that operates on the population of reads sharing the same identity-group membership, not on the broader specimen pool.

In the current implementation, this is computed in `_validate_identity_group` as `group_N = sum(len(subclusters[i]['read_ids']) for i in sorted_indices)`.

Why the identity-group sum is the right choice. The artifact hypothesis under test is “the candidate cluster is misclassified reads from its sibling.” The relevant population is the union of reads that could plausibly have ended up in either cluster — the identity group. Using the total specimen reads as N would inflate the denominator relative to the hypothesis under test, producing artificially small p^* values (and CER factors) and over-accepting weak variants. Using only the candidate cluster’s own size as N would conversely understate the denominator and over-reject. The identity-group sum corresponds exactly to the population over which the candidate is hypothesized to be a redistribution of reads from validated siblings.

Why not “total specimen reads”. A natural alternative is to use the total specimen input reads as N , on the conservative reasoning that any read in the specimen could in principle have appeared in this position. We considered and rejected this. The artifact null is not “any specimen read could have erred here” — which is what total specimen N models — but “among the reads belonging to this identity group, could redistribution have produced the candidate’s count.” Reads outside the group are part of unrelated identity groups and do not enter the artifact mechanism for this candidate.

5.2 Identity groups

An *identity group* is a connected component of clusters whose pairwise HP-normalized identity exceeds a threshold. Groups define the comparison set for pairwise CER: a candidate variant is tested against members of its own identity group, not against the entire specimen.

Construction. Groups are formed by union-find using `hp_normalized_pairwise_compare` as the distance function. The default threshold is 85% identity (15% adjusted-identity distance), tunable via `--group-identity`. Two clusters are joined if their pairwise identity exceeds the threshold; transitive closure forms groups.

Role in CER decisions. Within each identity group, clusters are ordered by size. The largest cluster (the *anchor*) auto-passes CER without comparison. Each smaller cluster is tested pairwise against all already-validated clusters in the group; the candidate’s CER factor is the *minimum* across all pairwise comparisons, corresponding to the “nearest plausible artifact source.” Acceptance requires the minimum factor to exceed the decision threshold.

Why groups, not any-to-any. The artifact hypothesis is that a small cluster represents reads misclassified from a similar cluster. This hypothesis only makes sense within a similarity group; clusters of distinct sequences cannot serve as artifact sources for each other. Restricting the comparison set to identity-group members both encodes this hypothesis and reduces compute by avoiding $O(n^2)$ comparisons across the specimen.

5.3 Pairwise CER comparison

Pairwise CER between candidate cluster A and validated reference cluster B proceeds as follows:

1. Compute pairwise differences $K = \text{count_variant_differences}(A_{\text{cons}}, B_{\text{cons}})$, where each substitution contributes 1 and each contiguous indel event contributes 1 (regardless of indel length).
2. For each of the K differing positions, classify the context and look up $q_{\text{ctx},i}$.
3. Compute the joint per-read error probability $\prod_i q_{\text{ctx},i}$.
4. Take $L = \max(|A_{\text{cons}}|, |B_{\text{cons}}|)$ as the consensus length input to n_{sites} (which subtracts HP-aggregated runs as described in Section 2.3).
5. Take N as the identity-group sum (Section 5.1) and M as the candidate cluster’s read count.
6. Solve Equation 2 for p^* at the configured α , then compute the CER factor.

The candidate’s reported CER factor is the minimum over all pairwise comparisons against validated cluster references. This conservative choice reflects the worst-case artifact source and ensures that if any nearby validated cluster could plausibly explain the candidate as artifact, the candidate is suppressed.

5.4 Phasing without statistical gating

Recursive MSA-based phasing in *speconsense* produces candidate clusters using only local frequency and count thresholds (`--min-variant-frequency`, `--min-variant-count`); CER is not evaluated during phasing. This is a deliberate design choice, not an oversight, and we intend to keep it that way under the context-aware framework.

Phasing as candidate generation. The role of phasing is to *partition* reads into candidate haplotype clusters based on positional allele patterns. Phasing answers the structural question “which reads co-vary at which positions,” producing a set of candidate clusters whose existence is a hypothesis to be tested downstream. The role of CER is to *evaluate* those candidates against the artifact null. Conflating generation and evaluation forces every phasing decision through the CER machinery, which is both expensive and structurally inappropriate: at the moment a phasing split is being considered, the candidate cluster has not yet had its consensus generated, so its K relative to siblings is unknown and the appropriate q_{ctx} values cannot be computed.

Why post-phasing CER is preferred. Three reasons:

1. **Mixed-context $K > 1$ requires consensus alignment.** The unified equation requires per-position $q_{\text{ctx},i}$ values for each of the K positions distinguishing the candidate from its sibling. These are known only after both consensus are generated and aligned, which happens post-phasing. A phasing-time gate would be limited to $K = 1$ evaluation against an arbitrary reference, losing the substantial sensitivity gains of multi-position evaluation (Section 2).
2. **Phasing’s local thresholds are fast and effective pre-filters.** The existing `--min-variant-count 5` and `--min-variant-frequency 0.10` thresholds eliminate the overwhelmingly majority of trivial splits (a single-read singleton at a random position) without any statistical machinery. CER’s contribution is discriminating among the remaining candidates, all of which have at least modest local support. Running CER at every phasing step would be largely wasted work.
3. **Identity-group N depends on phasing’s output.** The identity-group sum used as N in pairwise CER (Section 5.1) is determined by the set of clusters phasing produces and how they group by similarity. The hypothesis a phasing-time gate would test (“is this sub-split significant against *some* pool”) has no well-defined N until the full set of clusters is known.
4. **Cleaner primary consensus via noise excision.** Indiscriminate phasing followed by post-validation rejection lets non-significant variants be cleanly identified as their own *ns* (not significant) clusters rather than being absorbed back into a parent and manifesting as IUPAC ambiguity codes. With a phasing-time gate, reads that would have formed a non-significant split remain in their parent cluster, contributing to ambiguity at the affected positions and dirtying the parent’s consensus. The no-gate design separates “structurally a candidate variant” from “statistically meaningful,” so a primary consensus accumulates ambiguity only from positions where reads genuinely could not be phased apart at all—not from positions where phasing succeeded but the resulting cluster was too weakly supported to report. The non-significant reads remain visible in the output as *ns* clusters, available for inspection without polluting the significant consensus.

What about over-splitting? A reasonable concern is that indiscriminate phasing produces clusters that should never have been split, inflating downstream compute and adding noise. In practice, the local frequency and count thresholds do most of this work: a 10% / 5-read floor prevents single-read splits and cleanly catches obvious noise. Post-phasing identity-group construction (Section 5.2) collapses near-identical splits via union-find, and pairwise CER then suppresses weakly-supported candidates. The pipeline’s three-stage structure (local-threshold phasing → similarity-based grouping → statistical validation) cleanly separates concerns and matches each stage to the question it is structurally suited to answer.

5.5 HP detection and normalization

The existing pipeline applies HP-length normalization in two places:

1. During clustering identity computation (`_hp_normalized_pairwise_compare`), HP length differences within runs of length ≥ 6 contribute zero distance.
2. During MSA-based variant detection, HP length differences are blanket-discarded as “not real variants” regardless of length.

The first behavior is preserved: HP normalization in clustering distance is a separate concern from variant calling and need not change.

The second behavior changes when the context-aware framework is active. HP length variants at $L \leq 5$ become candidate variants subject to CER evaluation with the appropriate q_{ctx} . HP length variants at $L \geq 6$ continue to be blanket-discarded, following the recommendation of [2] §8.1.

Pipeline ordering. The MSA variant detector must be reordered so that HP length differences are surfaced as candidate variants (with their context tags) rather than dropped. The CER validation phase then evaluates these candidates with the same machinery as substitution variants. Implementation note: the existing `is_variant_position_with_composition` function tracks `homopolymer_composition` separately from raw base counts; this information feeds the context classifier directly.

Ambiguity handling. HP length variants that fail CER (factor < 2) should be absorbed into the parent cluster’s consensus as ambiguity, mirroring the behavior for failed substitution variants. Whether to encode HP length ambiguity as IUPAC codes, as the modal length, or as a special annotation is an open question; the current implementation uses the modal length without explicit ambiguity annotation. A revised approach could record HP length variation (e.g., `poly_A[5..6]` as a header annotation) for downstream inspection.

6 Metadata reproducibility

The CER computation depends on several inputs — N , M , K , n_{sites} , q_{ctx} , α — and a stable mapping from position to context. As empirical q_{ctx} values are revised (new basecaller versions, per-run estimation, improved error characterization), practitioners should be able to recompute CER factor and p-value from *stored* metadata, without re-running the pipeline. This section specifies the metadata schema that supports this.

6.1 Specimen metadata JSON

Each specimen produces a metadata JSON file at the standard output location. The current schema [3] contains parameters, input information, and primer settings; we extend it with CER reproduction data:

```
{
  "schema_version": "2.0",
  "sample_name": "BCFCQ-25-005",
  "timestamp": "2026-04-16T12:34:56Z",
  "parameters": {
    ...existing fields...,
  }
}
```

```

    "assumed_error_rate": 0.015,
    "significance_level": 1e-5,
    "qctx_table": "dorado-v5.0.0",
    "qctx_table_version": "20251201",
    "qctx_source": "shipped"
  },
  "specimen": {
    "total_input_reads": 1000,
    "consensus_length": 698,
    "n_sites": 587,
    "n_sites_breakdown": {
      "non_hp": 423,
      "hp_runs": 164
    }
  },
  "identity_groups": [
    {
      "group_id": "g1",
      "group_N": 942,
      "members": ["c1", "c2", "c3"]
    }
  ],
  "variants": [
    {
      "cluster_id": "c2",
      "identity_group": "g1",
      "anchor": false,
      "M": 42,
      "K": 2,
      "L": 698,
      "context_tags": ["non-hp-sub", "hp-l3-A-del1"],
      "q_ctx_per_position": [0.0059, 0.0213],
      "q_ctx_joint": 1.257e-4,
      "compared_against": "c1",
      "cer_pstar": 0.031,
      "cer_factor": 5.3,
      "p_value_at_qctx": 2.7e-9
    }
  ]
}

```

6.2 Recomputation workflow

Given a stored metadata JSON and a revised q_{ctx} table or significance level, recomputation proceeds:

1. For each variant, look up the new $q_{\text{ctx},i}$ values for the recorded `context_tags`.
2. Compute the new joint $q_{\text{ctx},\text{joint}} = \prod_i q_{\text{ctx},i}^{\text{new}}$.
3. Recompute p^* from $(N, M, n_{\text{sites}}, K, \alpha^{\text{new}})$.
4. The CER factor becomes $p^*/q_{\text{ctx},\text{joint}}^{\text{new}}$ where $q_{\text{ctx},\text{joint}}^{\text{new}}$ is the geometric mean of the new per-position rates raised to $1/K$, or equivalently a per-position factor based on the arithmetic structure preferred for the application.

5. P-value is $n_{\text{sites}} \cdot P(\text{Binom}(N, q_{\text{ctx,joint}}^{\text{new}}) \geq M)$.

The original N , M , and consensus-derived n_{sites} are immutable properties of the sequencing data; no upstream pipeline run is needed to revise the significance metric.

6.3 Schema versioning

The `schema_version` field allows downstream tools to detect missing fields and degrade gracefully. Version 1.0 is the pre-context-aware schema; version 2.0 introduces the CER fields in this paper. Tools reading version 1.0 output can fall back to the original raw- p^* interpretation; tools reading version 2.0 output have full reproducibility.

6.4 Why this is worth the bytes

The metadata addition adds $\sim 200\text{--}500$ bytes per variant. For a typical ITS amplicon study with 1000 specimens and an average of 3 variants each, the total overhead is on the order of 1–2 MB across the run. This is negligible relative to the FASTA and FASTQ output. The benefit — re-computation under revised assumptions without re-running — is substantial: as basecaller versions evolve and empirical error tables improve, every stored output remains scientifically valid without recomputation cost.

7 Validation

This section is a placeholder. Empirical content will be added once Phase 1 implementation generates data.

The planned validation reproduces the structure of the original CER validation [1] §8 but with the context-aware framework:

- Apply context-aware CER to the ONT98 dataset (837 specimens) and compare per-variant decisions against the existing raw- p^* implementation at default settings.
- Apply to run116 (127 specimens, herbarium type material) and quantify the impact of the cross-run q_{ctx} scaling factor identified in [2] §5.4 on accept/reject decisions.
- Sensitivity analysis: vary q_{ctx} by $\pm 50\%$ at each context class and quantify the fraction of decisions that flip. This characterizes the framework’s robustness to q_{ctx} misspecification under realistic uncertainty.
- HP-variant rescue: identify HP length variants that the current pipeline discards via blanket normalization but the context-aware framework would accept. Manual review of a sample to assess biological plausibility.
- Specimen-level outcomes: confirm that no specimens are lost (zero organism dropouts), consistent with the original CER validation.

Expected results: a moderate increase in HP-variant calls (mostly $L = 1\text{--}3$ variants currently discarded), a modest reduction in marginal substitution variants at HP-interface positions (where the empirical q_{ctx} is slightly lower than the current uniform assumption), and stable specimen-level counts.

8 Limitations

The framework as presented inherits the limitations of the underlying CER paper [1] and adds a few specific to the context-aware extension.

Cross-read error correlation. The binomial null assumes independence of errors across reads. Reads sharing the same PCR template can exhibit correlated errors (PCR-introduced substitutions, amplification bias); these are physically present in the library and not addressed by the CER framework, which establishes physical library presence rather than biological reality.

Within-read non-independence at HP positions. HP length errors are signal-level events affecting the entire run at once, not independent per-base errors [2] §3.4. The per-position q_{ctx} tabulated in the empirical analysis is a compact parameterization for cross-context comparison; the binomial model used here is correct because it operates across reads (independent draws from the sequencing process), not within a read. Practitioners should not interpret the per-position q_{ctx} for HP positions as a mechanistic rate.

Non-stationary error rates within a run. If error rates drift during a sequencing run (flow cell degradation, temperature variation), the binomial model’s identical-trials assumption is violated. Phase 2 per-run estimation partially mitigates this for between-run drift but cannot address within-run drift.

q_{ctx} provenance for novel basecaller versions. Shipped q_{ctx} tables are derived from empirical analysis of specific basecaller versions [2]. New basecaller releases require either re-characterization or use of the closest available table with explicit acknowledgment of the mismatch. Per-run estimation (Phase 2) addresses this but is not yet implemented.

Long HP runs ($L \geq 6$). The framework does not extend to $L \geq 6$ HP variants; these continue to be blanket-discarded as in the existing pipeline. The HP error analysis [2] §8.1 documents that accuracy degrades sharply at $L \geq 6$ and variant calling becomes unreliable; this is a property of the data, not the framework.

Biological variant priors. The framework treats all positions as having the same prior probability of carrying a real biological variant. In ITS, conserved regions (5.8S) and hypervariable regions (ITS1, ITS2) have very different prior variant densities. A Bayesian extension could incorporate position-specific priors; the current framework does not.

9 Conclusion

The CER framework [1] provides a transparent statistic for variant significance that is interpretable against platform error knowledge. Its extension to context-dependent error rates — demanded by the empirical characterization of HP errors [2] — requires changes to the inputs of the CER equation, not to its structure. The unified equation (Section 2) accommodates per-context error rates and mixed- context multi-position variants without abandoning the binomial framework.

The CER factor (Section 3) is the practical metric that recovers cross-variant comparability under context-aware q_{ctx} . It preserves the human-interpretable units that motivated the original framework, degrades linearly under realistic q_{ctx} uncertainty, and re-scales cleanly when the empirical error table is updated.

The implementation in speconsense (Section 5) integrates the framework with existing pipeline structure: the N denominator follows the conservative choice at each stage, identity groups define the pairwise comparison set, and HP variant detection is reordered so that $L \leq 5$ HP length differences become candidate variants subject to CER rather than being unconditionally discarded.

The metadata reproducibility design (Section 6) ensures that stored output can be reinterpreted under updated assumptions. As basecaller chemistry evolves and empirical error tables improve, no recomputation from raw reads is required — the pipeline’s conclusions remain adaptable to the moving target of nanopore error characterization.

Phase 2 work — per-run q_{ctx} estimation, an empirical-distribution model for HP variants that captures within-read signal-level dependencies, and incorporation of biological-variant priors — remains future work, scoped beyond the initial context-aware deployment.

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A Metadata JSON schema (full example)

A minimal but complete example showing all CER-related fields. Comments (prefixed with `//`) are illustrative and not part of the JSON.

```
{
  "schema_version": "2.0",
  "sample_name": "BCFCQ-25-005",
  "timestamp": "2026-04-16T12:34:56Z",
  "speconsense_version": "0.7.0",

  "parameters": {
    "algorithm": "mcl",
    "min_identity": 0.90,
    "presample_size": 1000,
    "assumed_error_rate": 0.015,
    "significance_level": 1e-5,
    "factor_threshold": 2.0,
    "qctx_table": "dorado-v5.0.0",
    "qctx_table_version": "20251201",
    "qctx_source": "shipped"
  },

  "input_file": "BCFCQ-25-005.fastq.gz",
  "specimen": {
    "total_input_reads": 1000,
    "consensus_length": 698,
    "n_sites": 587,
```

```

    "n_sites_breakdown": {
      "non_hp_positions": 423,
      "hp_runs_l2_plus": 164
    }
  },
  "identity_groups": [
    {
      "group_id": "g1",
      "group_N": 942,
      "members": ["c1", "c2", "c3"]
    },
    {
      "group_id": "g2",
      "group_N": 58,
      "members": ["c4"]
    }
  ],
  "variants": [
    {
      "cluster_id": "c1",
      "identity_group": "g1",
      "anchor": true,
      "size": 850,
      "ric": 100,
      "M": 850,
      "cer_pstar": null,
      "cer_factor": null,
      "note": "anchor: auto-passes CER"
    },
    {
      "cluster_id": "c2",
      "identity_group": "g1",
      "anchor": false,
      "size": 50,
      "ric": 50,
      "M": 50,
      "K": 1,
      "L": 698,
      "context_tags": ["non-hp-sub"],
      "q_ctx_per_position": [0.0059],
      "q_ctx_joint": 0.0059,
      "compared_against": "c1",
      "cer_pstar": 0.087,
      "cer_factor": 14.7,
      "p_value_at_qctx": 1.2e-15,
      "decision": "accept"
    },
    {
      "cluster_id": "c3",
      "identity_group": "g1",
      "anchor": false,
      "size": 42,

```

```

    "ric": 42,
    "M": 42,
    "K": 2,
    "L": 698,
    "context_tags": ["non-hp-sub", "hp-l3-A-del1"],
    "q_ctx_per_position": [0.0059, 0.0213],
    "q_ctx_joint": 1.257e-4,
    "compared_against": "c1",
    "cer_pstar": 0.031,
    "cer_factor": 246.6,
    "p_value_at_qctx": 4.1e-9,
    "decision": "accept"
  }
]
}

```

B Default q_{ctx} table for Dorado v5.0

Drawn from [2] Tables 1, 2, and 4. Values are filtered per-position error rates (after biological-variant filtering) for the ont98 dataset (R10.4.1, Dorado SUP v5.0.0). Per-(base, length, direction) values for HP length changes are derived from the supplementary distributions; the table below shows summary values.

Table 2: Default q_{ctx} values for Dorado v5.0 contexts (per-position total error rate). Context tag convention: **hp-l{L}-{base}** for HP length changes; direction-specific subtags (**-del1**, **-ins1**) override the aggregate when available.

Context tag	q_{ctx}	Source
non-hp-sub	0.0059	[2] Table 5 (ont98)
non-hp-indel	0.0108	sum of Del + Ins, [2] Table 5
hp-interface-sub	0.0059	treated as non-HP per [2] §4.3
hp-l1-A	0.0067	[2] Table 4
hp-l1-T	0.0067	[2] Table 4 (pooled)
hp-l1-C	0.0067	[2] Table 4 (pooled)
hp-l1-G	0.0067	[2] Table 4 (pooled)
hp-l2-A	0.0083	[2] Table 4
hp-l3-A	0.0097	[2] Table 4
hp-l4-A	0.0099	[2] Table 4
hp-l5-A	0.0113	[2] Table 4
hp-l5-G	0.0140	per-base, [2] Table 10
hp-l6+	N/A	blanket discard, [2] §8.1

The shipped Phase 1 table will include direction-specific subtags (**-del1**, **-ins1**, **-del2+**) and per-base values at all lengths; Table 2 shows summary values for illustration. The precise table is generated from the supplementary **hp_distributions.tsv** files of [2] via a build script that ships with speconsense.

References

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